WEEK 2 · DAY 3 · OVERVIEW

Z-Scores and Percentile Ranks



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ABOUT 45 MIN WITH EXERCISES / 9 SECTIONS / 4 EXERCISES

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The Desk Lead's Note

This morning's note says the 10-year is "near the top of its one-year range" and 2s10s is "about average." Make both exact for 2 October 2026: a z-score and a percentile rank over the last 252 market days, each defined in writing and tied to Excel, both added to marketdata with tests that prove they use no future data. Then write the two sentences for the note: where each sits, never where it goes.

Your Route Through Day 3

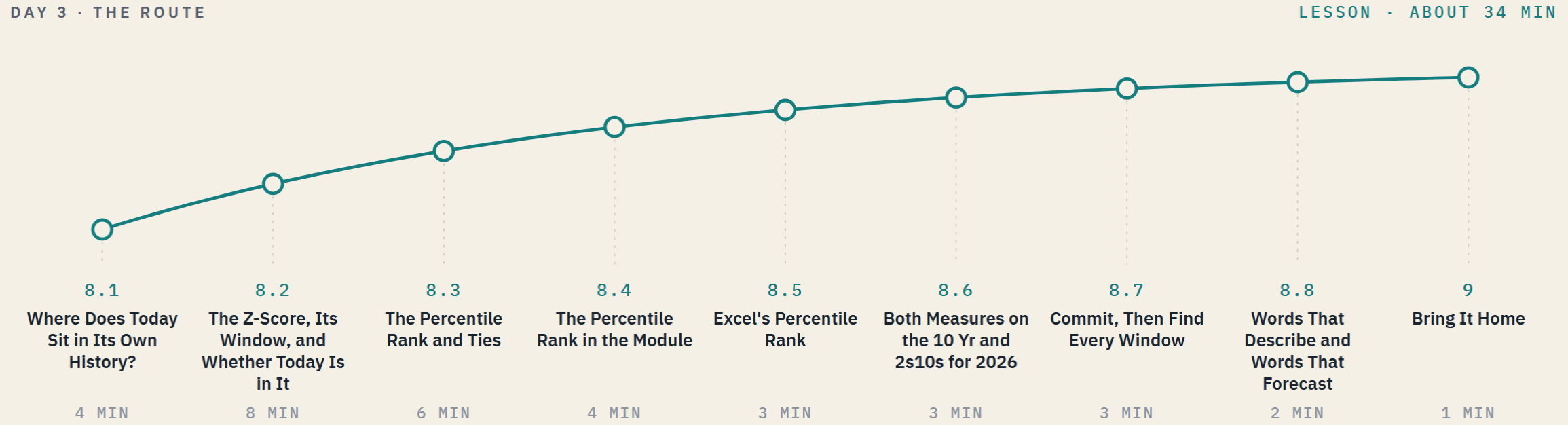


FIGURE 1 Nine sections, in order, with about how long each takes.

Time. About 34 minutes for the lesson, then about 12 for the four exercises.

8.1 Where Does Today Sit in Its Own History?

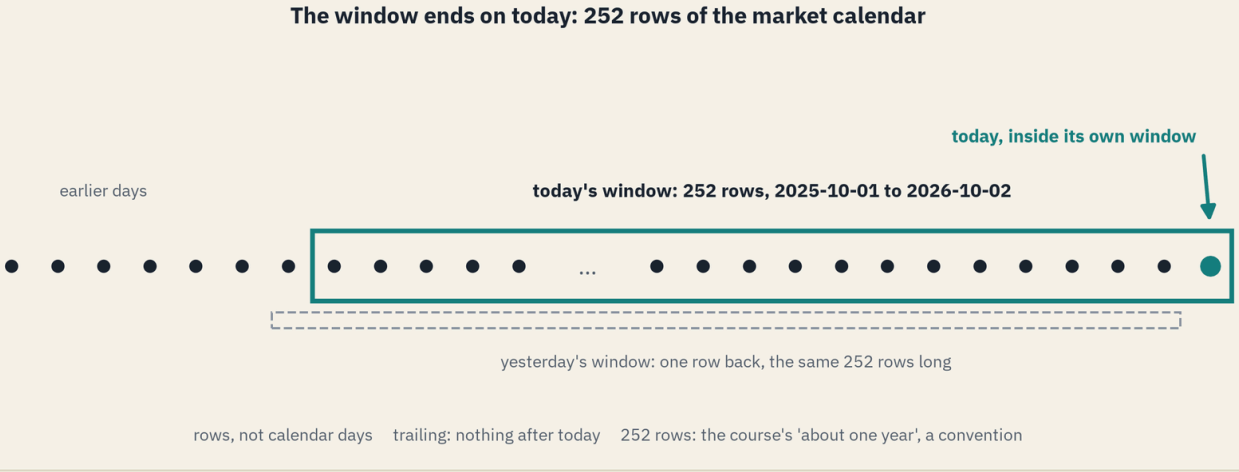


FIGURE 2 The last 252 rows, 2025-10-01 to 2026-10-02, form today's window; yesterday's is the same length, one row back.

EXCEL SIDE BY SIDE

JOB	EXCEL
Low and high	`=MIN(C3:C254),=MAX(C3:C254)`
The high's date	`=XLOOKUP(MAX(C3:C254),C3:C254,A3:A254)`
Distance from the mean, bp	`=(C254-AVERAGE(C3:C254))*100`

WHAT THE WINDOW HOLDS

```
2025-10-01 to 2026-10-02: today 5.28 percent
low 3.97 on 2025-10-22, high 5.29 on 2026-09-30, mean 4.3798
```

8.2 The Z-Score, Its Window, and Whether Today Is in It

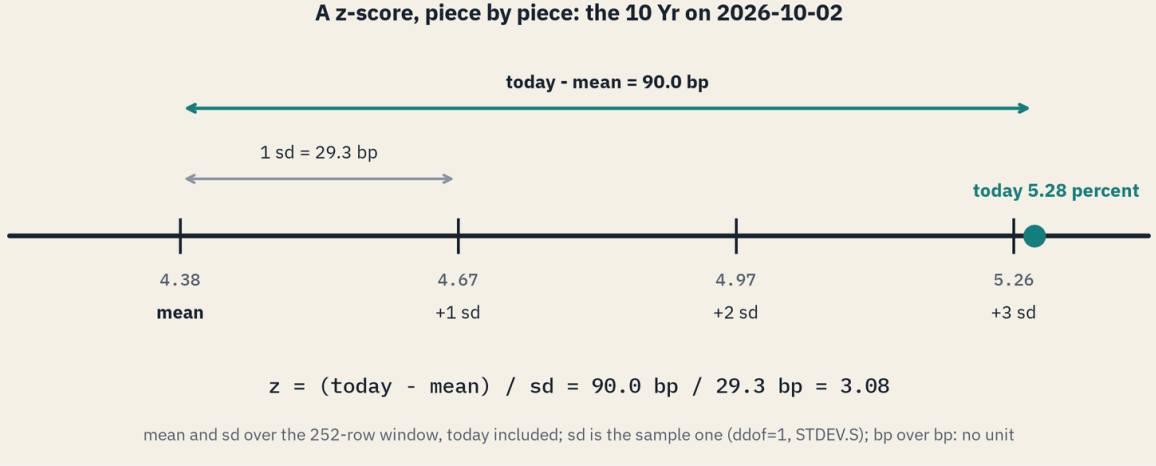


FIGURE 3 Today's 10 Yr, 5.28 percent, sits 90.0 bp above the window mean of 4.38; with sd 29.3 bp, the z-score is 3.08.

MODULE AGAINST HAND

```
z_ten_year = md.rolling_zscore(ten_year)

rolling_zscore today: 3.075234; by hand: 3.075234
251 NaN before the first full window
```

EXCEL SIDE BY SIDE

JOB	EXCEL
Z-score, today in	`=STANDARDIZE(C254,AVERAGE(C3:C254),STDEV.S(C3:C254))`
Today left out	`=STANDARDIZE(C254,AVERAGE(C2:C253),STDEV.S(C2:C253))`

`rolling_zscore` is Excel's `STANDARDIZE` on all 190 rows of 2026 for both series, and no future row reaches it.

8.3 The Percentile Rank and Ties

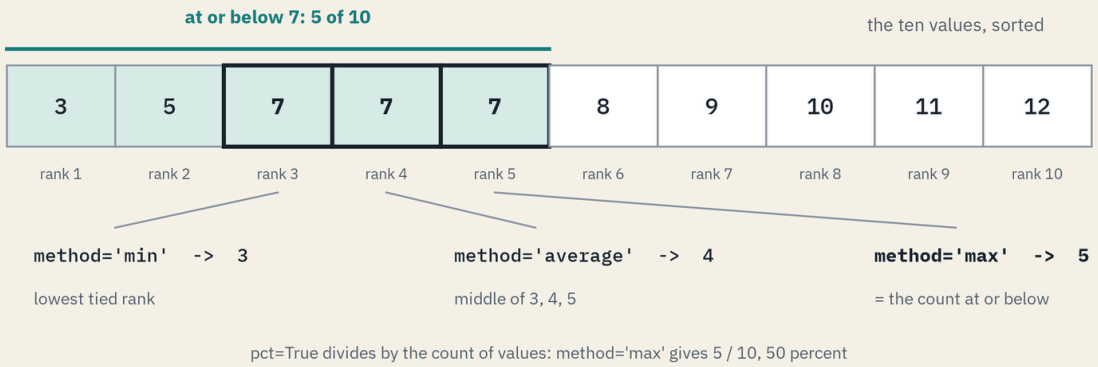


FIGURE 5 Of ten values with three 7s, five are at or below 7; only rank method max returns that count.

THREE TIE RULES

```
method='min': the 7s rank 3
method='average': the 7s rank 4
method='max': the 7s rank 5
```

EXCEL SIDE BY SIDE

JOB	EXCEL
Lowest tied rank	`=RANK.EQ(7,A1:A10,1)`
Average tied rank	`=RANK.AVG(7,A1:A10,1)`
At or below 7, percent	`=COUNTIF(A1:A10,"<="&7)/COUNT(A1:A10)*100`

8.4 The Percentile Rank in the Module

THE FUNCTION

```
def rolling_percentile_rank(series: pd.Series, window: int = 252) -> pd.Series:
    rounded = series.map(lambda value: float(f"{value:.15g}"))
    return rounded.rolling(window, min_periods=window).rank(pct=True, method="max") * 100
```

EXCEL SIDE BY SIDE

JOB	EXCEL
Percentile rank, 10 Yr	`=COUNTIF(C3:C254,"<="&C254)/COUNT(C3:C254)*100`
Percentile rank, 2s10s	`=COUNTIF(D3:D254,"<="&D254)/COUNT(D3:D254)*100`
The two 32s and 31	`=COUNTIF(A1:A3,"<="&A3)/COUNT(A1:A3)*100`

AGAINST COUNTIF

```
probe set, last value 8: 60 percent
10y: differs from COUNTIF on 0 of 190 rows
2s10s: differs from COUNTIF on 0 of 190 rows
```

8.5 Excel's PERCENTRANK.INC

EXCEL'S RULE, REBUILT

```
def percentrank_inc(values, x):
    below = (values < x).sum()
    return math.floor(below / (len(values) - 1) * 1000) / 1000
```

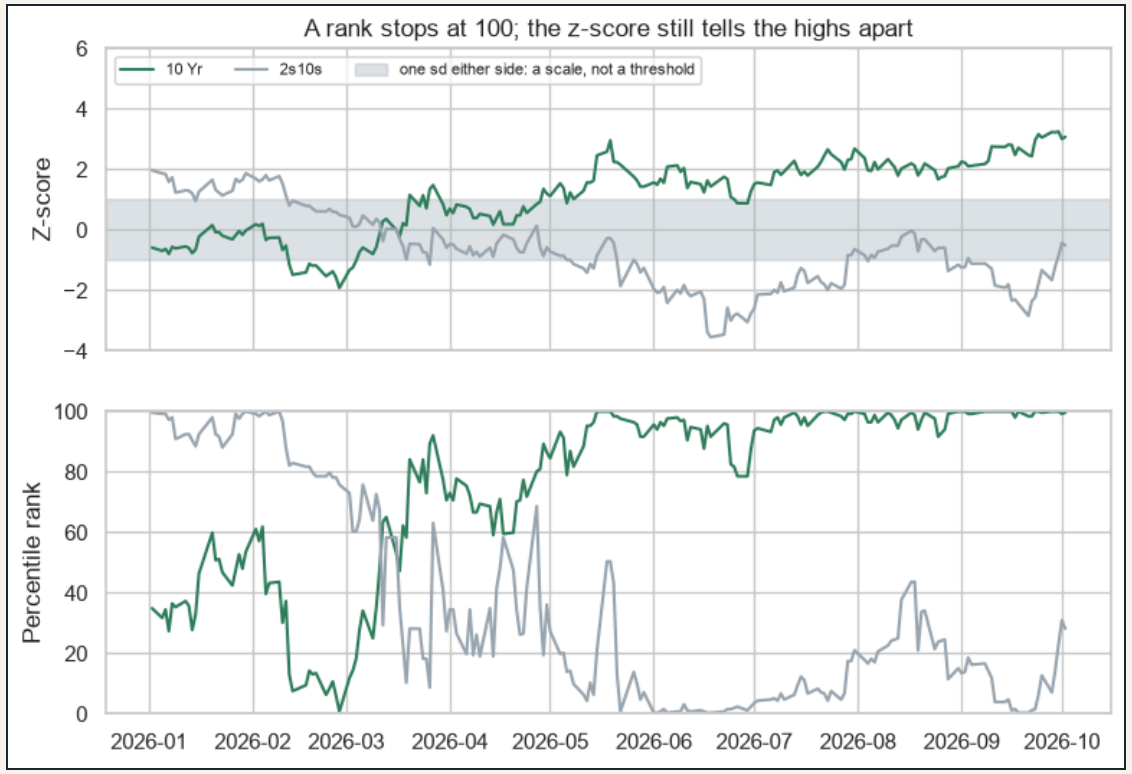
EXCEL SIDE BY SIDE

JOB	EXCEL
The ten values, 7	`=PERCENTRANK.INC(A1:A10,7)`
The ten values, 8	`=PERCENTRANK.INC(A1:A10,A10)`
The 10 Yr today	`=PERCENTRANK.INC(C3:C254,C254)`
2s10s today	`=PERCENTRANK.INC(D3:D254,D254)`

ON EVERY REFERENCE ROW

```
10y: PERCENTRANK.INC on 190 of 190 rows; up to 5.02 points from COUNTIF
2s10s: PERCENTRANK.INC on 190 of 190 rows; up to 10.13 points from COUNTIF
```

8.6 Both Measures on the 10 Yr and 2s10s for 2026



THE RANGE OF EACH

```
10y_z: -1.92 on 2026-02-27 to 3.25 on 2026-09-30
2s10s_z: -3.54 on 2026-06-18 to 1.97 on 2026-01-02
```

AGAINST EXCEL, EVERY ROW

```
10y: largest gap, z-score 1.9e-12; percentile rank 0.0e+00 points
2s10s: largest gap, z-score 7.3e-14; percentile rank 0.0e+00 points
```

On 2 October 2026 the 10 Yr sat 3.08 standard deviations above its one-year mean at the 99.6th percentile, and 2s10s 0.51 below at the 28.2nd.

8.7 Commit, Then Find Every Window

One search, every window: four rolling(lines, three defaults

```
398    def rolling_vol_bp(changes_bp, window: int = 63, annualise=False):        default 63 rows
...
413        vol = changes_bp.rolling(window, min_periods=window).std(ddof=1)
...
420    def rolling_zscore(series, window: int = 252):                            default 252 rows
...
434        mean = series.rolling(window, min_periods=window).mean()
435        std = series.rolling(window, min_periods=window).std(ddof=1)
...
441    def rolling_percentile_rank(series, window: int = 252):                   default 252 rows
...
459        return rounded.rolling(window, min_periods=window).rank(...) * 100

        git grep -n "rolling(" finds the green lines;  git grep -n "window: int = " finds the grey ones
```

FIGURE 8 git grep finds four rolling(lines and three window defaults, 63 and 252 rows, in marketdata.py.

WHAT IT SEARCHES

`git grep` searches only the files Git tracks, so the data files and **`bondmath.py`** stay out of it.

Before changing a default window, `git grep` finds every place one opens: four lines in three functions, defaults of 63 and 252 rows.

8.8 Words That Describe and Words That Forecast

THE TWO SENTENCES FOR THE NOTE

On 2026-10-02 the 10 Yr was 5.28 percent, 3.08 standard deviations above the mean of its last 252 market days, with 99.6 percent of those days at or below it. On 2026-10-02 2s10s was 45.0 bp, 0.51 standard deviations below the mean of its last 252 market days, with 28.2 percent of those days at or below it.

DESCRIBE, NEVER FORECAST

WORDS THAT DESCRIBE	WORDS THAT FORECAST OR ADVISE
3.08 standard deviations above the mean of its last 252 market days	stretched, extreme, overextended
99.6 percent of the window at or below today	overbought, oversold
the low and high of the window, with their dates	cheap, rich, attractive
up so many bp over the month (notebook 06)	due to revert, bound to come back

The note gets two sentences that say where each number sits and nothing about where it goes.

Copy the new functions into your own folder.

Bring It Home

IN A POWERSHELL TERMINAL

```
# the work folder: use the name your setup cell printed
$work = "$env:TEMP\pfi_course3_08_k4w7q2xr"
Set-Location "$HOME\projects\my_bondmath"
Copy-Item "$work\marketdata.py", "$work\test_marketdata.py" .
python -m pytest -q test_marketdata.py
git diff --stat
git add marketdata.py test_marketdata.py
git commit -m "Notebook 08: rolling_zscore and rolling_percentile_rank"
git grep -n "rolling(" -- marketdata.py
```

pytest shows 35 passed, and `git grep` the same four lines.

You Can Now

- Say where today sits in its own window as a distance in bp, a z-score, and a percentile rank.
- Compute a z-score over a trailing window that includes today, and show that leaving today out changes it.
- Compute a percentile rank that counts ties as at or below, comparing at 15 significant digits as **COUNTIF** does.
- Tell **PERCENTRANK.INC** from the share at or below, **COUNTIF** over **COUNT**.
- Prove a window function uses no future data, and find every window in a module with **git grep**.
- Describe a level without forecasting it.

THE NOTE ASKED	ANSWER
The 10 Yr	3.08 standard deviations above its one-year mean, 99.6th percentile
2s10s	0.51 below, 28.2nd percentile
Defined and tied to Excel	**STANDARDIZE** and **COUNTIF** over **COUNT**, 190 rows each, 35 passed

NEXT
WEEK 2 · DAY 4

Slopes and Butterflies

A curve's shape in two kinds of one-line arithmetic, slopes and butterflies, each with its sign written down: tied to Excel, checked against FRED, placed in its history, balanced by DV01 with your bondmath, and kept under Git.